

CS510 HW1

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Problem 1

1. TF Cosine Similarity

Vocab = woof, meow, squeak

D1 = $\langle 2, 1, 0 \rangle$

D2 = $\langle 2, 0, 1 \rangle$

$$\cos = \frac{4 + 0 + 0}{\sqrt{4 + 1}\sqrt{4 + 1}} = \frac{4}{5}$$

2. TF-IDF Cosine Similarity

(a) woof: $\log \frac{2}{2} = 0$

(b) meow: $\log \frac{2}{1} = 0.3$

(c) squeak: $\log \frac{2}{1} = 0.3$

TF-D1 = $\langle 2 \cdot 0, 1 \cdot 0.3, 0 \cdot 0.3 \rangle = \langle 0, 0.3, 0 \rangle$

TF-D2 = $\langle 2 \cdot 0, 0 \cdot 0.3, 1 \cdot 0.3 \rangle = \langle 0, 0, 0.3 \rangle$

$$\cos = \frac{0}{0.3} = 0$$

3. New TF-IDF Cosine Similarity

(a) woof: $\log \frac{3}{2} = 0.176$

(b) meow: $\log \frac{3}{2} = 0.176$

(c) squeak: $\log \frac{3}{2} = 0.176$

$$\text{TF-D1} = \langle 2 \cdot 0.176, 1 \cdot 0.176, 0 \cdot 0.176 \rangle = \langle 0.352, 0.176, 0 \rangle$$

$$\text{TF-D2} = \langle 2 \cdot 0.176, 0 \cdot 0.176, 1 \cdot 0.176 \rangle = \langle 0.352, 0, 0.176 \rangle$$

$$\cos = \frac{0.352^2}{0.155} = 0.8$$

Problem 2

1. 4.1

$$\begin{aligned} P(\text{sentence} \mid \text{pos}) &= P(\text{I} \mid \text{pos}) \cdot P(\text{always} \mid \text{pos}) \cdot P(\text{like} \mid \text{pos}) \cdot P(\text{foreign} \mid \text{pos}) \\ &\quad \cdot P(\text{films} \mid \text{pos}) \\ &= 0.09 * 0.07 * 0.29 * 0.04 * 0.08 \\ &= 0.0000058464 \end{aligned}$$

$$\begin{aligned} P(\text{sentence} \mid \text{neg}) &= P(\text{I} \mid \text{neg}) \cdot P(\text{always} \mid \text{neg}) \cdot P(\text{like} \mid \text{neg}) \cdot P(\text{foreign} \mid \text{neg}) \\ &\quad \cdot P(\text{films} \mid \text{neg}) \\ &= 0.016 * 0.06 * 0.06 * 0.15 * 0.11 \\ &= 0.0000009504 \end{aligned}$$

We could use log probabilities to get the same result, but reduce underflow. We see $P(\text{sentence} \mid \text{pos}) > P(\text{sentence} \mid \text{neg})$. Therefore, we determine that the sentence is classified as positive.

2. 4.2

Add-1 Smoothing:

$$P(w_i \mid c) = \frac{1 + \# \text{ of instances of } w_i \text{ in docs labeled c in D}}{\text{number of words in the vocabulary} + \# \text{ of docs labeled c in D}}$$

Vocabulary = [fun, couple, love, fast, furious, shoot, fly]

$$\begin{aligned}
P(\text{fun} \mid \text{comedy}) &= \frac{1+3}{7+2} = \frac{4}{9} \\
P(\text{couple} \mid \text{comedy}) &= \frac{1+2}{7+2} = \frac{3}{9} = \frac{1}{3} \\
P(\text{love} \mid \text{comedy}) &= \frac{1+2}{7+2} = \frac{3}{9} = \frac{1}{3} \\
P(\text{fast} \mid \text{comedy}) &= \frac{1+1}{7+2} = \frac{2}{9} \\
P(\text{furious} \mid \text{comedy}) &= \frac{1+0}{7+2} = \frac{1}{9} \\
P(\text{shoot} \mid \text{comedy}) &= \frac{1+0}{7+2} = \frac{1}{9} \\
P(\text{fly} \mid \text{comedy}) &= \frac{1+1}{7+2} = \frac{2}{9} \\
P(\text{fun} \mid \text{action}) &= \frac{1+1}{7+3} = \frac{2}{10} = \frac{1}{5} \\
P(\text{couple} \mid \text{action}) &= \frac{1+0}{7+3} = \frac{1}{10} \\
P(\text{love} \mid \text{action}) &= \frac{1+1}{7+3} = \frac{2}{10} = \frac{1}{5} \\
P(\text{fast} \mid \text{action}) &= \frac{1+2}{7+3} = \frac{3}{10} \\
P(\text{furious} \mid \text{action}) &= \frac{1+2}{7+3} = \frac{3}{10} \\
P(\text{shoot} \mid \text{action}) &= \frac{1+4}{7+3} = \frac{5}{10} = \frac{1}{2} \\
P(\text{shoot} \mid \text{fly}) &= \frac{1+1}{7+3} = \frac{2}{10} = \frac{1}{5}
\end{aligned}$$

Now we can determine our class for the sentence. We see $P(\text{action}) = \frac{3}{5}$

and $P(\text{comedy} = \frac{2}{5})$. Let $s = \text{fast, couple, shoot, fly}$. Therefore,

$$\begin{aligned}
 P(\text{comedy} \mid s) &= \frac{2}{5} \cdot \frac{2}{9} \cdot \frac{1}{3} \cdot \frac{1}{9} \cdot \frac{2}{9} \\
 &= \frac{8}{15 * 9 * 81} \\
 &= 0.00073159579 \\
 P(\text{action} \mid s) &= \frac{3}{5} \cdot \frac{3}{10} \cdot \frac{1}{10} \cdot \frac{1}{2} \cdot \frac{1}{5} \\
 &= \frac{9}{5 * 1000} \\
 &= 0.0018
 \end{aligned}$$

We see $P(\text{action} \mid s) > P(\text{comedy} \mid s)$. Therefore, we will classify this string as an action.

Problem 3

Code is in the zip file. Here are the cosine similarities from the non-pretrained Word2Vec model:

1. Top 10 similar words to 'rebellion':

- (a) decomposition: 0.9630
- (b) rattle: 0.9379
- (c) shocks: 0.9378
- (d) turtle: 0.9359
- (e) Marsh: 0.9356
- (f) Royal: 0.9327
- (g) drawings: 0.9311
- (h) wheat: 0.9303
- (i) clause: 0.9300
- (j) juice: 0.9295

2. Top 10 similar words to 'slave':

- (a) wider: 0.9360
- (b) alert: 0.9349
- (c) desire: 0.9315
- (d) wishes: 0.9290
- (e) marriage: 0.9279
- (f) pleasure: 0.9268
- (g) limit: 0.9266
- (h) charge: 0.9229
- (i) advance: 0.9223
- (j) price: 0.9222

Here are the cosine similarities from the pretrained Word2Vec model:

1. Top 10 similar words to 'rebellion':

- (a) revolt: 0.8357
- (b) insurrection: 0.7862
- (c) rebellions: 0.7244
- (d) uprising: 0.6939
- (e) revolts: 0.6581
- (f) mutiny: 0.6445
- (g) uprisings: 0.6043
- (h) rebellious: 0.5893
- (i) rebelled: 0.5875
- (j) insurrections: 0.5765

2. Top 10 similar words to 'slave':

- (a) slaves: 0.8382
- (b) slavery: 0.7260
- (c) enslaved: 0.7017
- (d) enslaved_Africans: 0.6437

- (e) slaver: 0.6239
- (f) slavers: 0.5992
- (g) Abraham_Lincoln_emancipation: 0.5779
- (h) slave_masters: 0.5764
- (i) abhorred_slavery: 0.5687
- (j) slaveholders: 0.5585